

Q1:

1. Draw the box plot.
2. Find a measure of central tendency. **Median = 45**
3. is the value "15" and outlier? Why? **[NO]**
 - $IQR = Q3 - Q1 = 60 - 40 = 20$
 - Lower Bound = $Q1 - 1.5 \times IQR = 40 - 30 = 10$
 - Upper Bound = $Q3 + 1.5 \times IQR = 60 + 30 = 90$
4. Direction Skewness: **right / positive skewed**
5. Calculate the coefficient of skewness

بالمعادلة هتبقى قيم تقريبية ف استخدمت بايثون

```
mean_value = np.mean(data)
median_value = np.median(data)
std_value = np.std(data, ddof=0) |
skewness = (3 * (mean_value - median_value)) / std_value
```

- Mean: **55**
- Median: **45**
- Standard Deviation: **22.47**
- Skewness Coefficient: **1.33**

Comment: **right-skewness is confirmed due to the positive value.**

6. Is it a difficult exam? Why? **YES**, because:
 - The median is 45 (**below the midpoint**).

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
data = [30, 33, 37, 40, 45, 60, 72, 80, 98]
```

```
plt.figure(figsize=(6, 4))
sns.boxplot(data=data)
```

```
plt.xlabel("Statistics Scores")
plt.title("Box Plot Draw-Task")
#plt.grid(True)
```

```
plt.show()
```

